

do42_classification_fundamentals _ADDENDUM

ADDENDUM — content to merge into the existing deck, not a standalone presentation.

Addendum for do42_classification_fundamentals. pdf

This is not a new deck — it is content to insert into the existing **Mastering Classification Fundamentals** presentation (`d042_classification_fundamentals.pdf`), right after the current decision-tree section (`cla_dtree`). It adds the alternate `rpart` backend for the same tree topic.

The Same Tree, a Different Backend — `cla_rpart`

CART-style classification tree using the `rpart` backend, instead of the tree implementation behind `cla_dtree` .

```
model <- cla_rpart("Species")
model <- fit(model, iris_train)
```

```
prediction <- predict(model, iris_test)
evaluate(model, iris_test$Species, prediction)
```

- **Didactic goal:** keep the multiclass classification line of experiment fixed and change only the tree backend — a controlled comparison of wrapper behavior, not a new modeling idea.
- Same configure → fit → predict → evaluate rhythm as `cla_dtree`.
- Useful talking point: when two wrappers implement "the same" algorithm family, what actually differs is engine defaults and available tuning knobs, not the experiment shape.

Reference: Breiman, L., Friedman, J., Olshen, R., and Stone, C. (1984). *Classification and Regression Trees*.

Suggested Placement

Insert as a short closing section titled "**Alternate Backend: rpart**" right after the existing decision-tree slides, before the deck moves on to the next classifier family. Frame it explicitly as *"same idea, different engine"* rather than a new technique, so it doesn't compete with the deck's main narrative arc.